Phase 8 Completion Report: Governance Artifacts

1. Phase Overview

Phase 8 of the Banking Data Governance & Data Quality Pipeline focused on creating manual governance artifacts that connect the completed technical pipeline to enterprise data governance concepts.

The purpose of this phase was to practice Collibra-style governance thinking without introducing a governance platform yet. Phase 8 did not install or configure OpenMetadata. OpenMetadata remains assigned to Phase 9.

Phase 8 was a documentation and governance artifact phase. It did not change source schema logic, ingestion logic, PySpark processing logic, curated model logic, SQL validation logic, or PySpark data quality validation logic.

The completed pipeline before Phase 8 already included:

PostgreSQL banking source system
→ Python ingestion layer
→ raw Parquet zone
→ PySpark standardized processing layer
→ standardized Parquet zone
→ curated warehouse fact and dimension layer
→ DuckDB SQL data quality validation layer
→ PySpark data quality validation layer

Phase 8 added a governance view on top of those completed layers.

2. Phase 8 Objective

The objective of Phase 8 was:

To build manual Collibra-style governance artifacts that define business terms, Critical Data Elements, data assets, source-to-target mappings, lineage, data quality rule references, stewardship issue examples, and governance operating rules for the existing banking data pipeline.

This phase helped connect technical data engineering work to governance concepts such as:

- Business glossary
- Data catalog
- Data domains
- Data ownership
- Data stewardship
- Critical Data Elements
- Data classification

- Source-to-target mapping
- Data lineage
- Data quality rule catalog
- Stewardship issue tracking
- Governance policy

3. Phase 8 Scope

Phase 8 was split into two parts:

- Phase 8A: Governance Foundation
- Phase 8B: Catalog, Lineage, DQ Rule Catalog, and Stewardship Artifacts

This split kept the scope disciplined and prevented the phase from becoming too large.

In Scope

Phase 8 included:

- Manual governance Markdown files
- Manual governance YAML files
- Business glossary terms
- Critical Data Elements
- Governance policy
- Data catalog entries
- Source-to-target mappings
- Lineage mappings
- Data quality rule catalog
- Sample stewardship issue records
- Functional owners and stewards
- Classifications such as PII, Confidential, Internal, and Critical Data Element
- Mapping governance artifacts to existing technical layers

Out of Scope

The following items were intentionally excluded from Phase 8:

- OpenMetadata setup
- OpenMetadata ingestion configurations

- Docker setup for governance tools
- Agile/JIRA simulation files
- New SQL validation scripts
- Changes to the DuckDB SQL validation runner
- Changes to PySpark DQ validation logic
- New Python automation scripts
- New PySpark transformations
- New curated models
- Runtime output generation
- Any changes to pipeline execution logic

This scope control was important because Phase 8 was meant to document and govern the existing pipeline, not modify how the pipeline runs.

4. Phase 8A: Governance Foundation

Phase 8A Goal

The goal of Phase 8A was to create the foundational governance artifacts for the project.

Phase 8A focused on:

- Governance folder documentation
- Governance policy
- Business glossary
- Critical Data Element definitions

Phase 8A Files Created

The following files were created:

- governance/README.md
- governance/governance_policy.md
- governance/glossary/business_terms.yml
- governance/metadata/critical_data_elements.yml

Phase 8A Work Completed

Phase 8A created the governance foundation for the project.

The governance README explains the purpose of the governance folder, the difference between Phase 8A and Phase 8B, and confirms that OpenMetadata belongs to Phase 9.

The governance policy defines how governance should operate in the project. It covers data domains, data owners, data stewards, data quality engineer responsibilities, Critical Data Element selection, data classifications, data quality expectations, lineage expectations, stewardship issue handling, and change control.

The business glossary defines important banking terms used in the project. These terms connect business language to physical fields and curated warehouse assets.

The Critical Data Elements file identifies the most important governed fields in the banking pipeline and maps them to classifications, owners, stewards, source fields, standardized fields, curated fields, related assets, and data quality rule categories.

Phase 8A Business Glossary Coverage

Phase 8A included glossary terms such as:

- Customer
- Account
- Branch
- Product
- Transaction
- Transaction Amount
- Transaction Date
- Transaction Type
- Account Balance
- Customer Risk Score
- Risk Band
- Critical Data Element

These terms help translate technical pipeline fields into governance-friendly business language.

Phase 8A CDE Coverage

Phase 8A defined 15 Critical Data Elements:

- customer_id
- customer_number
- email

- date_of_birth
- account_id
- account_number
- account_status
- current_balance
- transaction_id
- transaction_reference
- transaction_amount
- transaction_date
- transaction_type
- risk_score
- risk_band

Each CDE was mapped to business meaning, domain, classification, owner, steward, related technical fields, related assets, and relevant data quality checks.

Phase 8A Commit

Phase 8A was committed with:

2d82a68 Phase 8A - Add governance foundation artifacts

5. Phase 8B: Catalog, Lineage, DQ Rule Catalog, and Stewardship Artifacts

Phase 8B Goal

The goal of Phase 8B was to connect the Phase 8A governance foundation to actual technical assets, mappings, lineage paths, data quality rule categories, and stewardship examples.

Phase 8B Files Created

The following files were created:

- governance/catalog/data_assets.yml
- governance/metadata/source_to_target_mapping.yml
- governance/lineage/lineage_mapping.yml
- governance/data_quality/dq_rules.yml
- governance/stewardship/data_quality_issues.yml

Phase 8B Work Completed

Phase 8B created the technical governance mapping layer.

The data catalog file documents the main assets in the pipeline, including PostgreSQL source tables, raw Parquet assets, standardized Parquet assets, curated warehouse models, SQL DQ result assets, and PySpark DQ result assets.

The source-to-target mapping file documents how key source tables and fields flow into curated models.

The lineage mapping file documents asset-level and field-level lineage across the pipeline.

The data quality rule catalog documents the rule categories already implemented in Phase 6 and Phase 7.

The stewardship issue file provides sample governance issue records. These are clearly marked as sample records and are not actual current failures.

Phase 8B Source-to-Target Mapping Coverage

Phase 8B mapped the following major paths:

- customers to dim_customer
- accounts to dim_account
- branches to dim_branch
- products to dim_product
- transactions and accounts to fact_transaction
- customer_risk_scores to fact_customer_risk_score
- accounts and transactions to fact_daily_account_balance

These mappings support governance traceability from operational source data to curated warehouse outputs.

Phase 8B Lineage Coverage

Phase 8B documented lineage across the following flow:

PostgreSQL source
→ raw Parquet
→ standardized Parquet
→ curated warehouse model
→ SQL DQ results
→ PySpark DQ results

Field-level lineage examples were included for:

- customer_id
- account_id
- transaction_id

- transaction_amount
- transaction_date
- risk_score
- risk_band

This gives practical experience with how enterprise governance tools represent technical lineage.

Phase 8B Data Quality Rule Catalog Coverage

The DQ rule catalog references the validation categories already built in Phase 6 and Phase 7:

- null_check
- duplicate_key_check
- accepted_value_check
- range_check
- referential_integrity_check
- source_to_target_reconciliation
- control_total_check
- exception_record_review

No new SQL or PySpark validation logic was added in Phase 8B. The catalog only documents and references existing validation categories.

Phase 8B Stewardship Issue Examples

Phase 8B created sample stewardship issue records to show how governance teams would track and review data quality findings.

The records include fields such as:

- issue_id
- title
- status
- severity
- related_asset
- related_cde
- detected_by_rule
- assigned_steward

- business_impact
- remediation_plan
- notes

These examples support interview practice around data stewardship, issue triage, accountability, and remediation workflows.

Phase 8B Commit

Phase 8B was committed with:

5513cb3 Phase 8B - Add catalog lineage DQ and stewardship artifacts

6. Phase 8 Validation

Phase 8 validation confirmed the following:

- Phase 8A files were created successfully.
- Phase 8B files were created successfully.
- YAML files were readable.
- Business glossary terms were created.
- 15 CDEs were defined.
- Approved DQ categories were referenced.
- Source-to-target mappings were created.
- Lineage examples were created.
- Stewardship records were clearly marked as sample records.
- No pipeline code was changed.
- No SQL validation logic was changed.
- No PySpark validation logic was changed.
- No OpenMetadata tracked files were added.
- No Agile/JIRA tracked files were added.
- Runtime folders were not committed.
- Git working tree was clean.

One validation note: an initial folder-absence check failed because empty, untracked openmetadata/ and agile/ directories already existed locally. This was not a blocker because those directories were not modified and no tracked files were added under them.

7. Final Git History

After tagging Phase 8, the latest Git history was:

5513cb3 (HEAD -> main, tag: phase-8-complete) Phase 8B - Add catalog lineage DQ and stewardship artifacts
2d82a68 Phase 8A - Add governance foundation artifacts
59bc5f5 (tag: phase-7-complete) Phase 7B - Add PySpark DQ rule validation and exceptions
f0a4abf Phase 7A - Add PySpark DQ profiling foundation
1830481 (tag: phase-6-complete) Phase 6B - Add SQL reconciliation and exception checks

The following phase tags exist:

- phase-1-complete
- phase-2-complete
- phase-3-complete
- phase-4-complete
- phase-5-complete
- phase-6-complete
- phase-7-complete
- phase-8-complete

8. Final Phase 8 Status

Phase 8 is complete.

Final status:

- Phase 8A governance foundation: Complete
- Phase 8B governance mapping artifacts: Complete
- Business glossary: Complete
- Governance policy: Complete
- Critical Data Elements: Complete
- Data catalog YAML: Complete
- Source-to-target mapping YAML: Complete
- Lineage mapping YAML: Complete
- Data quality rule catalog: Complete
- Sample stewardship issue records: Complete
- Pipeline logic unchanged: Complete

- Runtime outputs ignored: Complete
- Git commits completed: Complete
- Phase tag created: Complete

9. Skills Practiced in Phase 8

Phase 8 helped practice the following skills:

- Data governance documentation
- Business glossary design
- Critical Data Element identification
- Data catalog design
- Source-to-target mapping
- Data lineage documentation
- Data quality rule cataloging
- Data classification
- Data ownership and stewardship concepts
- Governance policy writing
- Stewardship issue tracking
- Connecting business terms to technical fields
- Connecting CDEs to SQL and PySpark DQ checks
- Explaining governance concepts using a real data pipeline
- SDLC-style phased delivery
- Git commit and tagging discipline

10. Governance Relevance

Phase 8 is important because enterprise data governance is not only about running data pipelines. Governance also requires clear definitions, ownership, metadata, quality controls, lineage, and stewardship processes.

This phase connected the technical banking pipeline to governance practice in the following ways:

- Business terms were defined.
- Critical Data Elements were identified.
- CDEs were mapped to source, standardized, and curated fields.

- Data assets were cataloged.
- Source-to-target mappings were documented.
- Lineage was documented from source to curated and DQ result layers.
- DQ rule categories were cataloged.
- Sample stewardship issues were created.
- Governance roles and responsibilities were documented.
- OpenMetadata was intentionally left for the next phase.

This is the manual governance thinking behind tools such as Collibra and OpenMetadata.

11. Interview Explanation

A clear interview explanation for Phase 8 would be:

"In Phase 8, I built manual governance artifacts for my banking data governance pipeline. The earlier phases already created the technical flow from PostgreSQL source tables through Python ingestion, raw Parquet, standardized PySpark outputs, curated warehouse models, SQL validation, and PySpark data quality validation. In Phase 8, I created the governance layer on top of that technical pipeline.

I split the work into two parts. In Phase 8A, I created the governance foundation, including a governance policy, business glossary, and Critical Data Element definitions. I defined business terms such as Customer, Account, Transaction, Transaction Amount, Account Balance, Customer Risk Score, and Risk Band. I also identified CDEs such as customer_id, account_id, transaction_id, transaction_amount, transaction_date, risk_score, and risk_band, and mapped them to classifications, owners, stewards, source fields, curated fields, and data quality rule categories.

In Phase 8B, I created catalog, lineage, source-to-target mapping, DQ rule catalog, and stewardship issue artifacts. These artifacts documented how source tables such as customers, accounts, transactions, and customer_risk_scores flow into curated models such as dim_customer, dim_account, fact_transaction, and fact_customer_risk_score. I also documented lineage from PostgreSQL source to raw Parquet, standardized Parquet, curated models, SQL DQ results, and PySpark DQ results.

This phase helped me practice how governance teams connect business terms, CDEs, catalog assets, lineage, DQ rules, and stewardship issue tracking to real data engineering pipelines. I intentionally kept the phase documentation-only and did not change SQL, PySpark, ingestion, or modeling logic."

12. Phase 8 Conclusion

Phase 8 successfully created the manual governance artifact layer for the Banking Data Governance & Data Quality Pipeline.

The project now has a complete flow from source generation through governance documentation:

PostgreSQL banking source system
→ Python ingestion layer

→ raw Parquet zone
→ PySpark standardized processing layer
→ standardized Parquet zone
→ curated warehouse fact and dimension layer
→ DuckDB SQL data quality validation layer
→ PySpark data quality validation layer
→ manual governance artifacts

Final Phase 8 result:

- Phase 8A completed and committed.
- Phase 8B completed and committed.
- Phase 8 tagged as phase-8-complete.
- Git working tree is clean.
- No pipeline logic was changed.
- Governance artifacts now connect the technical pipeline to Collibra-style concepts.

The project is now ready for Phase 9: OpenMetadata Integration.